

AMMAR IRFAN

Monash ECSE Student | Bridging Engineering, Innovation & Community Impact

CONTACT

Clayton, VIC 3168

+61 450 315 041

✉ ammarbinirfan2003@gmail.com

🌐 LinkedIn: [amm-irfan-2bc](https://www.linkedin.com/in/amm-irfan-2bc)

🌐 Portfolio: <https://den-nu.vercel.app/>

🌐 GitHub: [am-cell](https://github.com/am-cell)

EDUCATION

Bachelor of Electrical and Computer Systems Engineering

Monash University, Malaysia

Oct 2023 – Feb 2024

Monash University, Melbourne, Australia

Feb 2024 – Expected June 2027

TECHNICAL SKILLS

Programming and HDL: Python, C, C++, Verilog, MATLAB, Assembly (MARS)

Embedded Hardware: STM32, Arduino, FPGA (Intel Quartus), SPI

Communication

Design and Simulation: LTSpice, SolidWorks

Core Competencies : Embedded Systems, Digital Logic, CPU

Architecture, Signal Processing,

Analogue circuits, Parallel Processing

CERTIFICATIONS &

AWARDS

Future MedTech Leaders Program

D K and G Anderson Award

Monash Energy Ideathon Innovation award

Monash High Achiever Award (HAA)

PROFILE

Electrical and Computer Systems Engineering student at Monash University with hands-on experience across embedded systems, FPGA design, and biomedical signal processing. Currently working on the research and prototyping of a portable hearing-loss screening device at the Monash Institute of Medical Engineering in collaboration with Ear Science Institute. Comfortable across the stack – HDL, microcontrollers, digital design and AI integration – with a growing focus on health-tech and human-centered engineering.

EXPERIENCE

Research Support Officer | Monash Medical Institute & Ear Science Institute June 2026 – Present

- Contributing to the research and prototyping of a portable hearing-loss screener, working across both hardware development and study design
- Working across circuit/system design and clinical research processes to translate engineering concepts into a usable diagnostic tool
- Collaborating with clinical and research staff to align device requirements with real-world screening use cases

Technical Intern | Maxme Nov 2025 – Feb 2026

- Developed AI avatar video content using Heygen AI.
- Researched APIs and automation tools to improve internal processes and workflow efficiency

Technical Intern | Aftab Engineering Nov 2024 – Jan 2025

- Assisted with process automation using Omron PLCs to improve operational efficiency

ENGINEERING PROJECTS

Final Year Project – Violet Light as a Replacement for Blue Light in MIMO VLC Systems

- Researched the feasibility of violet light replacing blue light in MIMO Visible Light Communication systems, due to blue light's significantly greater effects on sleep, arrhythmia, and mental health
- Compared violet vs. blue light across communication performance and biological safety criteria to assess feasibility as a system-level design choice

FPGA System with Camera Interfacing, User Input, and Haptic Feedback

- Designed an FPGA system with camera interfacing, user inputs, and ADXL345-based haptic feedback

- Built in parallel processing on the FPGA to handle multiple input streams simultaneously
- Used SPI communication to interface the camera and the VGA output

Audio Processing on STM32

- Implemented real-time audio sampling and filtering at 44.1 kHz using UART+DMA on STM32 microcontrollers and streamed to a PC

Analog True RMS Converter

- Designed and built a discrete analog true RMS-to-DC converter using operational amplifiers and passive components
- Analysed RMS accuracy for non-sinusoidal waveforms through LTSpice simulation and experimental validation

FPGA CPU Design (Verilog)

- Designed a 16-bit RISC CPU including ALU, control unit, and memory interfaces using Verilog

Biomedical Signal Processing (Tremor Analysis)

- Research project for wearable tech focused on analyzing tremors in patients using real-time signal processing techniques
- Submitted to James Dyson Award & selected for Monash UROP research

Smart Wi-Fi and gesture Robot & IoT Home System

- Developed a gesture-controlled Wi-Fi robot using ESP8266 and MPU6050 sensors
- Designed an IoT home automation system for lighting and sensor control with custom user interfaces